Clovis SFEIR

+33 6 95 83 95 34 · clovis.sfeir.cs@gmail.com · github.com/sfeirc · linkedin.com/in/clovis-sfeir-388936240 · clooooo.com · French (Native) – English (C1)

Profile

IMT Atlantique engineering student and AI engineer apprentice (Infotel AI Lab) who builds and ships ML/NLP systems in Python and Rust. Built a from-scratch BM25 retrieval engine (**MRR 0.944**), a from-scratch LoRA fine-tuning engine, a governed MLOps platform with drift-gated model promotion, and an HNSW vector index (**recall@10 0.993**); cut lookup time by **40%** with a RAG pipeline during two AI internships.

Education

IMT Atlantique – Engineering Degree in Software Engineering (Apprenticeship) *Sep 2025 – Jun 2028*
Nantes, France *Algorithms & Data Structures, OS, Computer Networks, Distributed Systems, Cloud Computing, AI*

Bachelor in Computer Science *Sep 2022 – Jun 2025*
La Rochelle, France ***Class rank: 1/85*** *| Software Engineering, Databases, Cybersecurity*

HRT Macro Placement Challenge 2026 (Hudson River Trading/Partcl) — **24th/124** (top 20%); hybrid analytical placer: force-directed global placement with timing-aware legalization; proxy cost **1.095** beats SA baseline (2.125) and RePlAce (1.458); field includes Stanford, Princeton, ETH Zürich.

Projects

Engineering Docs RAG *Python, BM25, information retrieval*

- From-scratch Okapi BM25 retrieval engine (chunking + ranking) for engineering documents (datasheets, HAZOP-style procedures); pluggable extractive/LLM generator interface, zero paid API calls in tests/CI/demo by design.
- Hand-derived BM25 scoring verified to 9 significant figures; measured **precision@1 0.90**, recall@5 1.00, **MRR 0.944** on a 40-question labeled set; ~3,330 docs/s ingest; 53 tests, zero runtime dependencies.

LoRA Fine-Tuning Engine *Python, PyTorch, from-scratch transformer*

- Hand-written decoder-only transformer and LoRA layer on raw PyTorch autograd (no PEFT/Trainer); LoRA math checked against hand-computed values, plus a test that frozen base weights stay non-trainable.
- Measured 3-way comparison (no fine-tune / full / LoRA r=8): LoRA trains only **3.85%** of parameters (24.6K vs 614.8K) and forgets less (generic val. loss 2.20 vs 3.08 for full fine-tuning) but adapts less to the new domain (2.33 vs 0.80); 13 tests.

Sentinel MLOps Platform *Python, FastAPI, drift monitoring*

- Governed model lifecycle on synthetic credit-risk data: one versioned feature pipeline shared by training and serving (equivalence-tested), FastAPI serving, and a registry with **seven promotion gates** plus a separate attributed approval step.
- Fixed-bin PSI and two-sample KS drift detection: in the deterministic demo (1,100 requests) a population shift raised critical drift, the retrained challenger was rejected by policy, and the champion kept serving with **zero request errors**.

HNSW Vector Search *Rust, approximate nearest neighbours*

- From-scratch HNSW graph index plus an exact brute-force baseline: the retrieval core behind vector databases and dense-retrieval RAG.
- Measured **recall@10 0.9933** at 20,000 vectors (dim 32, 3 seeds, M=16, ef_search 100) against exact search; recall degradation at larger scale is measured and documented; 40 tests.

IIoT Predictive Maintenance *Rust, tokio, anomaly detection (EWMA/CUSUM)*

- Async telemetry pipeline (`tokio` mpsc) fuses EWMA + CUSUM control charts across 3 sensor channels to detect developing bearing-wear signatures, with linear-trend remaining-useful-life estimation.
- Sensor-fusion tuning (2-of-3 channels + widened control limits) cut false positives from a 0.41–0.50 precision baseline to **0/10 across 10 seeds**; precision **1.0000**, recall 0.9642, F1 0.9818, 43-sample latency; **1.6 M samples/s**; 26 tests, 0 failures.

Work Experience

Infotel *Nantes, France*
AI Engineer Apprentice – Lab IA *Sep 2025 – Jun 2028*

- Built a Rust-based autonomous software-delivery harness that takes a complete specification through architecture synthesis (subsystems, interface contracts), epic/slice decomposition, and TDD-first generation to a tested, deployable application end-to-end; automated compile+test quality gates with iterative self-repair, independent code-review pass, and cross-slice conflict detection before delivery; 928 Rust tests; validated end-to-end producing a fully working, 100%-test-passing application directly from a single specification.

Excelia Group *La Rochelle, France*
Software Developer Intern / AI Software Engineer Intern *Two internships: Jan–Mar 2025 & May–Jul 2024*

- RAG pipeline (chunking, OpenAI embeddings, flat FAISS, cross-encoder re-ranking): **40% reduction** in lookup time vs. keyword-search; cut p95 latency 30–40% via embedding cache + asyncio parallelisation; Prometheus/Grafana, Docker, GCP.
- Real-time interview simulator (Next.js/React, GPT-4o, ElevenLabs TTS): streaming responses, scenario-branching state machine, A/B prompt-scoring harness.

Technical Skills

Languages	C++20, Python, Rust, TypeScript, SQL
Perf / Systems	OpenMP, cache-aware layout, acquire/release atomics, ICS/SCADA protocol parsing (Modbus, IEC-104)
ML / AI	PyTorch, LoRA fine-tuning, BM25/HNSW/FAISS retrieval, RAG, PSI/KS drift monitoring, EWMA/CUSUM, LightGBM/XGB/CatBoost
Infrastructure	Docker, tokio async, Prometheus/Grafana, GitHub Actions CI, GCP, Redis, ZeroMQ
Frameworks	FastAPI, Flask, Node.js/Express, React/Next.js, REST/gRPC